Type MS · Apply digestive enzyme action to an unfamiliar situation

Answer 1 of 6 · Question 9 part a.ii · 3 marks

2020 JAN · Paper 02 · find part a.ii in this mark scheme extract

Number				
9	(a) (i)	Any three from: • amino group removed/broken down(1) • in liver(1) • by deamination(1) • to produce urea (1)	Allow description of amino group e.g. nitrogenous group	3
	(ii)	Any three from: • less protein broken down by protease(1) • to amino acids 1) • in stomach/small intestine(1) • reference to reduced amount of phenylalanine/amino acids (in low protein diet)(1)		3
	(iii)	• reduced number of amino acids available(1) • during translation(1) • on ribosomes(1) • incorrect/less protein formed(1)	Allow less translation Allow less protein synthesis	3
	(b) (i)	heterozygous/Nn(1)		1
	(ii)	100% (1)		1
	(iii)	• both parents carry recessive allele/allele for PKU/heterozygous(1) • person 3 has inherited one recessive allele from each parent(1) • person 3 has PKU/homozygous recessive(1)	Reject any reference to sex-linked	3

Total for Question 9 = 14 marks

Type MS · Apply digestive enzyme action to an unfamiliar situation

Answer 2 of 6 · Question 7 part c.i · 2 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part c.i in this mark scheme extract

7(a)	<table><tr><th>Description</th><th>Letter</th></tr><tr><td>stores sperm</td><td>G; (1)</td></tr><tr><td>produces sperm</td><td>H; (1)</td></tr><tr><td>produces fluid to help sperm move</td><td>C; (1)</td></tr><tr><td>contains cells dividing by meiosis</td><td>H; (1)</td></tr></table>		**(4)**
Question Number	Answer	Additional guidance	Mark
7(b)(i)	progesterone; increases in level;		**(2)**
Question Number	Answer	Additional guidance	Mark
7(b)(ii)	egg not released until day 14; sperms may not still be active (at day 14)/sperm can't fuse with ovum/sperm cant fertilise ovum;	reference to sperm must be made	**(2)**
Question Number	Answer	Additional guidance	Mark
7(c)(i)	will be digested/broken down (in duodenum/ ileum/intestine); by lipase; can't be absorbed into blood in active form;		**(2)**

Question Number	Answer	Additional guidance	Mark
7(c)(ii)	no chance of becoming pregnant for a seven day period/menstruation occurring/lining being shed;		**(1)**

Total 11 Marks

Type MS · Apply digestive enzyme action to an unfamiliar situation

Answer 2 of 6 · Question 7 part c.i · 2 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part c.i in this mark scheme extract

Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Apply digestive enzyme action to an unfamiliar situation

Answer 3 of 6 · Question 7 part a.i · 2 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

Number					
7	(a)	(i)	• allows protein to stay longer in stomach/gut (1) • gives time for full digestion (1)		2
		(ii)	• diet largely/entirely milk when young (1) • changes to solids when older (1)		2
		(iii)	• biuret test (1) • add reagent to solid (1) • blue to lilac/pale purple (1)		3
	(b)	(i)	• temperature (1) • volume of milk (1) • volume/concentration of chymosin (1)		3
		(ii)	$\frac{238 + 232 + 241 + 229}{4}$ (1) = 235 (1)		2
		(iii)	not included when calculating mean (1)		1
		(iv)	bubbles of different sizes/volume of CO_2 variable (1)		1

Total for this question = 14 marks

Type MS · Apply digestive enzyme action to an unfamiliar situation

Answer 3 of 6 · Question 7 part a.i · 2 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Apply digestive enzyme action to an unfamiliar situation

Answer 4 of 6 · Question 5 part b · 1 mark

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

5 (a)	• two monosaccharides; • joined together;		2
(b)	molecule too large;		1
(c) (i)	• transcription (of lactase gene) does not occur; • RNA is not produced/synthesised; • lactase/protein/enzyme not produced;		3
(ii)	• lactose can't be digested; • into components to be absorbed;		2
(d)	• they eat more dairy produce; • being able to digest it would give them an advantage;		2
(e)	• water passes from cells; • into gut; • diarrhoea causes poor absorption of vitamin D and Ca^{2+}; • as they are water soluble;		4

Total 14 marks

Type MS · Apply digestive enzyme action to an unfamiliar situation

Answer 4 of 6 · Question 5 part b · 1 mark

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Apply digestive enzyme action to an unfamiliar situation

Answer 5 of 6 · Question 5 part c.ii · 2 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part c.ii in this mark scheme extract

5 (a)	• two monosaccharides; • joined together;		2
(b)	molecule too large;		1
(c) (i)	• transcription (of lactase gene) does not occur; • RNA is not produced/synthesised; • lactase/protein/enzyme not produced;		3
(ii)	• lactose can't be digested; • into components to be absorbed;		2
(d)	• they eat more dairy produce; • being able to digest it would give them an advantage;		2
(e)	• water passes from cells; • into gut; • diarrhoea causes poor absorption of vitamin D and Ca^{2+}; • as they are water soluble;		4

Total 14 marks

Type MS · Apply digestive enzyme action to an unfamiliar situation

Answer 5 of 6 · Question 5 part c.ii · 2 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part c.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Apply digestive enzyme action to an unfamiliar situation

Answer 6 of 6 · Question 7 part a.ii · 2 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

Number					
7	(a)	(i)	• allows protein to stay longer in stomach/gut (1) • gives time for full digestion (1)		2
		(ii)	• diet largely/entirely milk when young (1) • changes to solids when older (1)		2
		(iii)	• biuret test (1) • add reagent to solid (1) • blue to lilac/pale purple (1)		3
	(b)	(i)	• temperature (1) • volume of milk (1) • volume/concentration of chymosin (1)		3
		(ii)	$\frac{238 + 232 + 241 + 229}{4}$ (1) = 235 (1)		2
		(iii)	not included when calculating mean (1)		1
		(iv)	bubbles of different sizes/volume of CO_2 variable (1)		1

Total for this question = 14 marks

Type MS · Apply digestive enzyme action to an unfamiliar situation

Answer 6 of 6 · Question 7 part a.ii · 2 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks